SZABIST
University

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ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 01

Submitted To: **Mr. Muhammad Azhar**

Student Name: **M. Mateen Bashir**

Reg. Number: **23108125**

1) List at least 10 features / functional requirements of your project

Prepare a scenario/case study for your own project idea and provide feature list relevant to the proposed scenario.

ADD KEYWORDS FOR SPECIFIC SUBJECT AND PROJECT

2) Provide a feasibility study

Prepare a table mapping each requirement to available Tools/resources/APIs/packages/libraries that would help you implement.

Sr.	Requirements	Resources

3) Technical developer guide

Choose any three features from your feature list and provide a detailed, step-by-step technical guide on how to implement them.

E.g. Step by step procedure to install a required library, to configure an API, and to implement a requirement by using specific Tools/resources/APIs/packages/libraries.

4) Implementation

Using any ONE of the available tools/technology stack, implement the three selected features inside a small working project.

Submission Requirements:

- Push your completed mini project to your own GitHub repository.
- Include a short README file describing the added features and navigation flow.
- Prepare screenshots of your project code from android studio, output from emulator or separate device, and your git repository along with feasibility study to include in your submission document.

Solution:

Real-Time QR Code Scanner System

Scenario

A university library wants to scan student QR IDs for entry logging.

Keywords

Computer Vision, QR Detection, Barcode Scanning, OpenCV QRCodeDetector, Real-Time Processing.

Functional Requirements

1. Capture webcam feed
2. Detect QR codes
3. Decode QR data
4. Display decoded text
5. Log entry
6. Timestamp record
7. Save log file
8. Show bounding box
9. Prevent duplicate scan
10. Exit with ESC

Feasibility

Sr	Requirement	Tools
1	QR detection	QRCodeDetector()
2	Decode data	detectAndDecode()
3	Logging	File handling
4	Timestamp	datetime
5	Display	putText()

Technical Feasibility

OpenCV provides built-in QR detection.

Economic Feasibility

No paid QR API required.

Operational Feasibility

Simple library entry automation.

Technical Guide

Feature 1: QR Detection

detector = cv2.QRCodeDetector()

Feature 2: Decode

data,bbox,_ = detector.detectAndDecode(frame)

Feature 3: Logging

with open("qr_log.txt","a") as f:

f.write(data+" - "+time+"\n")

```
import cv2
from datetime import datetime

detector = cv2.QRCodeDetector()
cap = cv2.VideoCapture(0)

scanned = set()

while True:
    ret, frame = cap.read()
    data,bbox,_ = detector.detectAndDecode(frame)

    if data:
        if data not in scanned:
            scanned.add(data)
            now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
            with open("qr_log.txt","a") as f:
                f.write(data+" - "+now+"\n")
```

```
        cv2.putText(frame,data,(50,50),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    1,(0,255,0),2)

    cv2.imshow("QR Scanner",frame)

    if cv2.waitKey(1)==27:
        break

cap.release()
cv2.destroyAllWindows()
```

QRScannerSystem Public

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23108125-boop Add files via upload 4268a47 · now 3 Commits

README.md	Update README.md	1 minute ago
app.py	Add files via upload	now
qr_log.txt	Add files via upload	now

README

Real-Time QR Code Scanner

Description

Scans QR codes using OpenCV and logs entry data.

Features

- QR detection
- Decode QR content
- Entry logging
- Duplicate prevention
- Timestamp storage
- Real-time webcam display

Requirements

pip install opencv-python

Run

python app.py

Output

- Webcam window with QR detection
- qr_log.txt containing scanned data